



暨南大学  
JINAN UNIVERSITY

# University Physics

Department of Electronic Engineering  
College of Information Science and Technology

授课教师：杨希娅 副教授

办公地点：番禺校区 李国华楼（校医院）二楼 2012

邮箱：xiyayang@jnu.edu.cn



# Electromagnetic Waves

- 33-1 The Nature of Light
- 33-2 Reflection and Refraction
- 33-3 Total Internal Reflection
- 33-4 Dispersion
- 33-5 Polarization
- 33-6 Scattering of Light
- 33-7 Huygens' Principle



# 33-1 The Nature of Light

- Isaac Newton and most scientists before him proposed that light is composed **of streams of particles**, based on the phenomenon of rectilinear propagation of light.
- Around 1665, the wave nature of light was gradually discovered. By the 19th century, the scientific community was convinced that light is a type of wave. In 1873, the crowning achievement of James Clerk Maxwell was to show that **a beam of light is a traveling wave of electric and magnetic fields—an electromagnetic wave**—and thus that optics, the study of visible light, is a branch of electromagnetism.

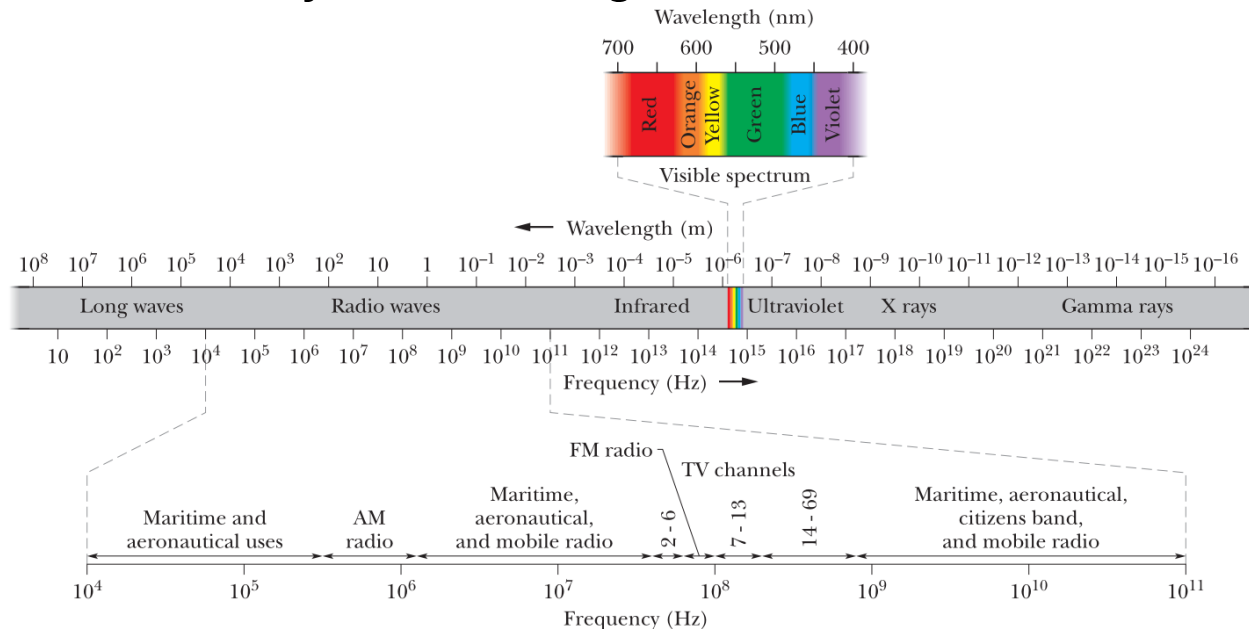


Figure 33-1 The electromagnetic spectrum.



# 33-1 The Nature of Light

- Yet, the understanding of light did not end there. The emission and absorption of light demonstrated its particle properties, which are associated with **photons**. By the 1930s, with the development of quantum electrodynamics, people came to realize **that light possesses a dual nature of both wave and particle**.
- The measured light speed in 1983:  $c = 2.99792458 \times 10^8 \text{ m/s}$

*Wave Speed.* From Eq. 16-13, we know that the speed of the wave is  $\omega/k$ . However, because this is an electromagnetic wave, its speed (in vacuum) is given the symbol  $c$  rather than  $v$ . In the next section you will see that  $c$  has the value

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \quad (\text{wave speed}), \quad (33-3)$$

which is about  $3.0 \times 10^8 \text{ m/s}$ . In other words,

- In 1983, the International Organization of Weights and Measures stipulated that the speed of light in a vacuum is exactly equal to this value, and based on this, the unit of length "meter" was defined.

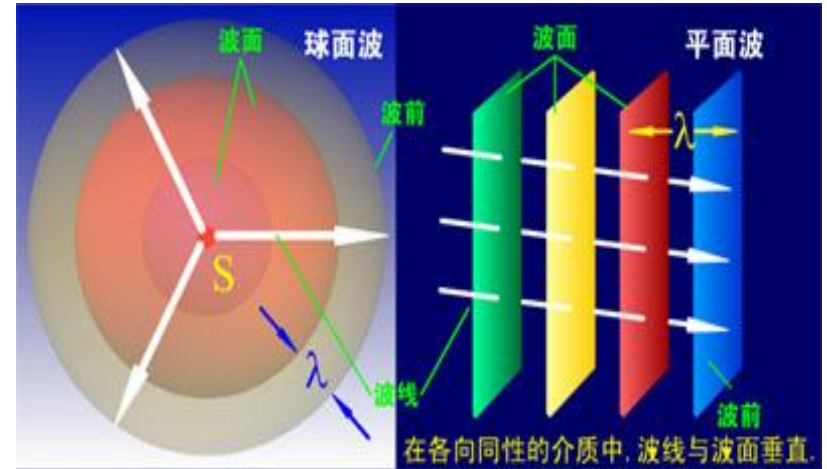


# 33-1 The Nature of Light

Several physical quantities:

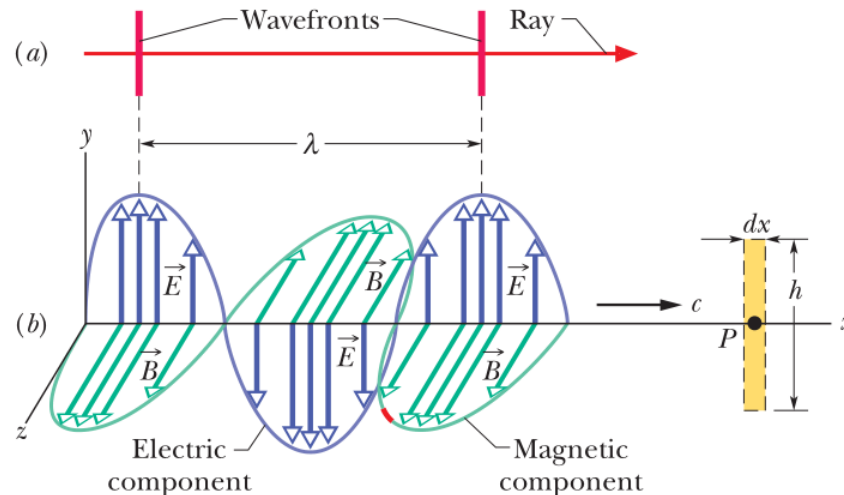
## 1. Wave fronts: Equiphas surface

Light waves are classified according to the shape of their wavefronts, including the plane waves, spherical waves, and cylindrical waves.



## 2. Rays: A straight line along the direction of light propagation.

In homogeneous isotropic media, light rays are always perpendicular to the wavefront.



# 33-2 Reflection and Refraction

- **Geometric optics**: Using ray patterns to study the properties of light.
- **Physical optics**: Using wave theory to study the properties of light.

## 33-2 Reflection and Refraction

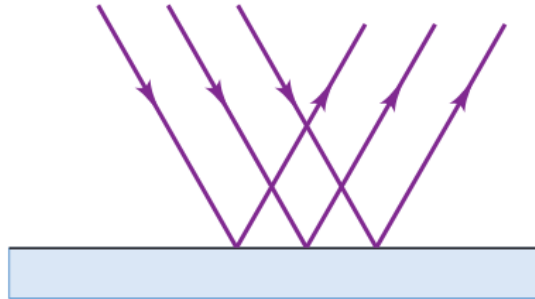
Light undergoes **reflection** and **refraction** at the interface between two media. Part of the **incident rays** are reflected back to the original medium and become **reflected rays**, while the other part is refracted into the second medium and becomes **refracted rays**.

- **Specular reflection**: Unidirectional reflection on a smooth interface
- **Diffuse reflection**: Reflection in all directions on a rough surface.
- **Index of refraction**:  $n = c / v$

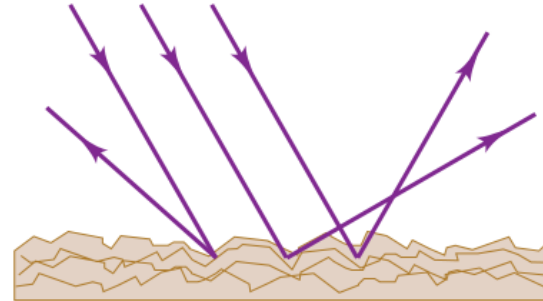


# 33-2 Reflection and Refraction

(a) Specular reflection



(b) Diffuse reflection



## In the right Figure:

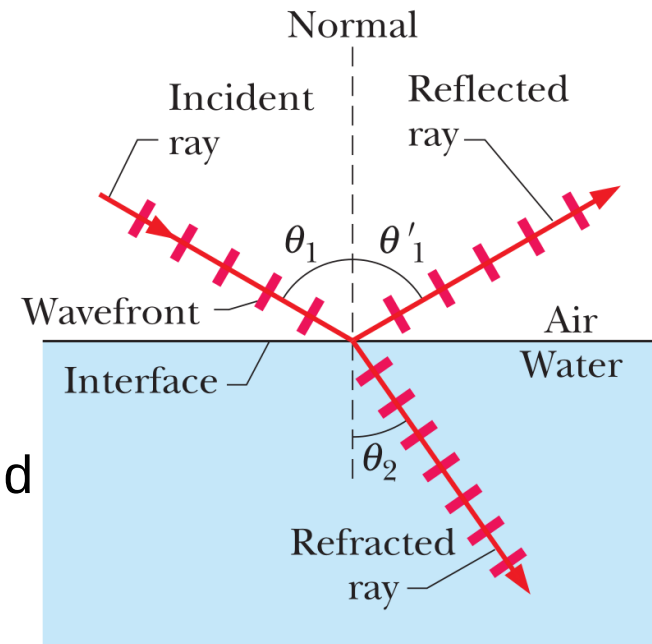
The **angle of incidence** is  $\theta_1$

The **angle of reflection** is  $\theta'_1$

The **angle of refraction** is  $\theta_2$

All measured *relative to the normal*.

The plane containing the incident ray and the normal is the *plane of incidence*, which is in the plane of the page.



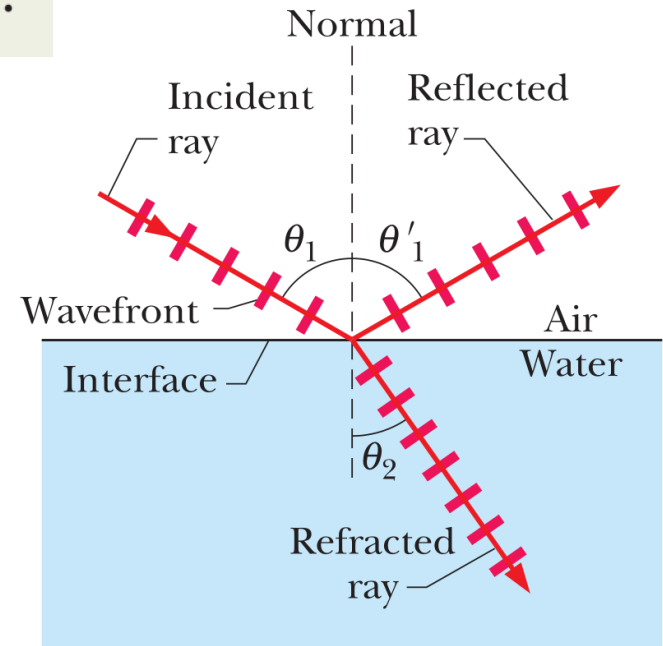
## 33-2 Reflection and Refraction

- **Law of reflection:** A reflected ray lies in the plane of incidence and has an angle of reflection equal to the angle of incidence (both relative to the normal).

$$\theta'_1 = \theta_1 \quad (\text{reflection}).$$

- **Law of refraction (Snell's Law):** A refracted ray lies in the plane of incidence and has an angle of refraction  $\theta_2$  that is related to the angle of incidence  $\theta_1$  by:

$$n_2 \sin \theta_2 = n_1 \sin \theta_1 \quad (\text{refraction}).$$



Where  $n$  is the index of refraction,  $n_1$  is the index of refraction of air, and  $n_2$  is the index of refraction of water.



# 33-2 Reflection and Refraction

## *Law of refraction:*

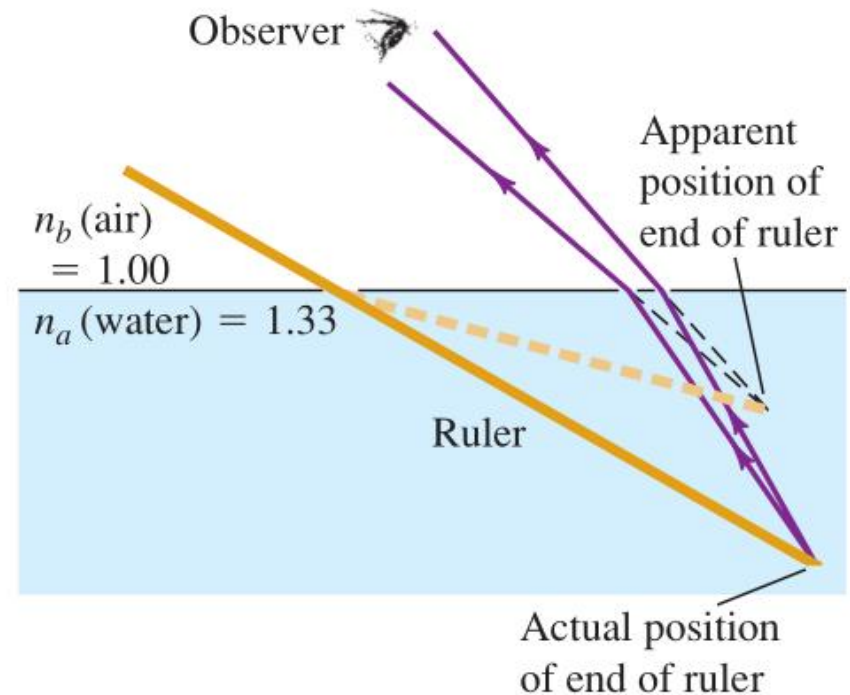
$$n_2 \sin \theta_2 = n_1 \sin \theta_1 \quad (\text{refraction}).$$

**33.9** (a) This ruler is actually straight, but it appears to bend at the surface of the water. (b) Light rays from any submerged object bend away from the normal when they emerge into the air. As seen by an observer above the surface of the water, the object appears to be much closer to the surface than it actually is.

(a) A straight ruler half-immersed in water



(b) Why the ruler appears bent



## 33-2 Reflection and Refraction

- Light rays are refracted and deflected **toward the medium with the higher refractive index** (light rays incident perpendicularly do not change direction after refraction).
- The paths of reflected and refracted light rays are **reversible**.
- Although the laws of reflection and refraction are experimental laws, they can be derived from Maxwell's equations, and the intensity, amplitude, and polarization of reflected and refracted light can also be derived.
- The phenomenon where the refractive index changes with the frequency of light waves is called **dispersion**.  $n = n(f)$
- The frequency of light remains constant during propagation, but the wavelength/speed changes depending on the medium.

$$\lambda = \lambda_0 / n$$



# Example 1

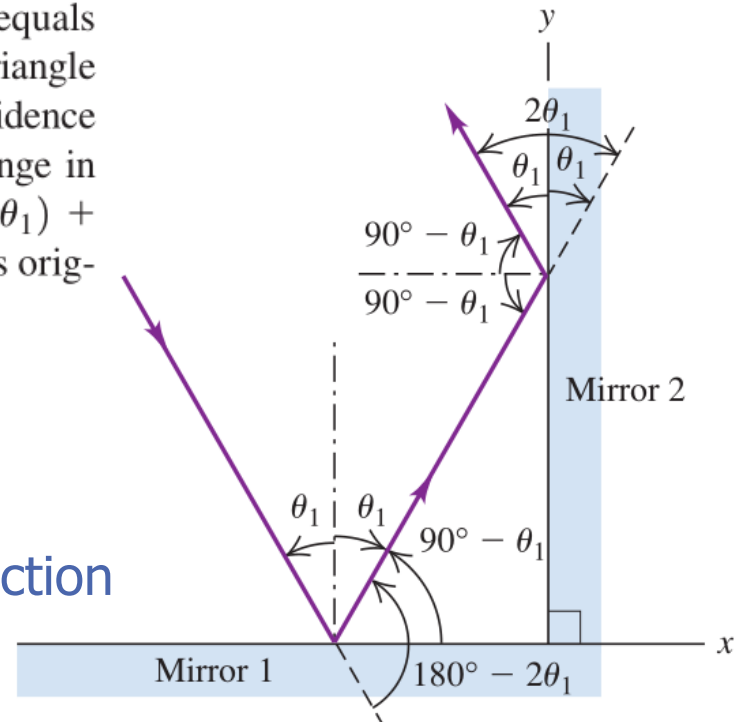
## A twice-reflected ray

Two mirrors are perpendicular to each other. A ray traveling in a plane perpendicular to both mirrors is reflected from one mirror, then the other. What is the ray's final direction relative to its original direction?

**EXECUTE:** For mirror 1 the angle of incidence is  $\theta_1$ , and this equals the angle of reflection. The sum of interior angles in the triangle shown in the figure is  $180^\circ$ , so we see that the angles of incidence and reflection for mirror 2 are both  $90^\circ - \theta_1$ . The total change in direction of the ray after both reflections is therefore  $2(90^\circ - \theta_1) + 2\theta_1 = 180^\circ$ . That is, the ray's final direction is opposite to its original direction.

$$2(90^\circ - \theta_1) + 2\theta_1 = 180^\circ$$

Parallel, and opposite to its original direction

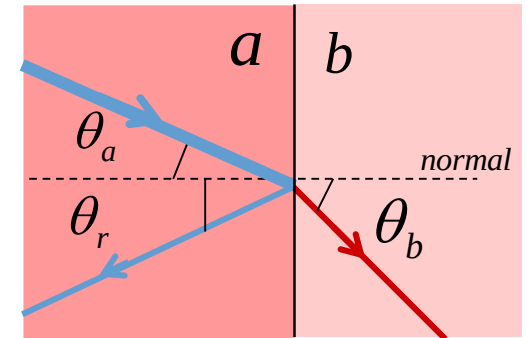


# 33-3 Total Internal Reflection

When light propagates from a denser medium ( $n$  is large) to a less dense medium ( $n$  is small):  $n_a > n_b$

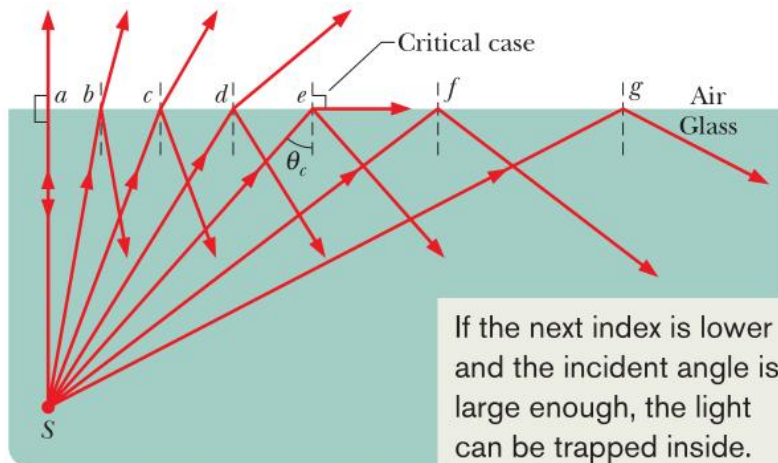
$$\theta_a < \theta_b, \quad \theta_b = 90^\circ \Rightarrow \theta_a = \theta_{crit}$$

$$\sin \theta_{crit} = n_b / n_a$$



**Total internal reflection:**  $\theta_a \geq \theta_{crit}$

The phenomenon where light that has no refraction is completely reflected back is called *total internal reflection*.



(b)



Ken Kay/Fundamental Photographs



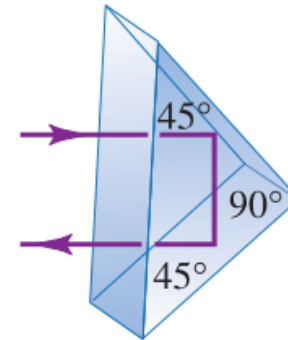
# 33-3 Total Internal Reflection

## Applications of Total Internal Reflection

Total internal reflection finds numerous uses in optical technology. As an example, consider glass with index of refraction  $n = 1.52$ . If light propagating within this glass encounters a glass–air interface, the critical angle is

$$\sin \theta_{\text{crit}} = \frac{1}{1.52} = 0.658 \quad \theta_{\text{crit}} = 41.1^\circ$$

The light will be *totally reflected* if it strikes the glass–air surface at an angle of  $41.1^\circ$  or larger. Because the critical angle is slightly smaller than  $45^\circ$ , it is possible to use a prism with angles of  $45^\circ - 45^\circ - 90^\circ$  as a totally reflecting surface.

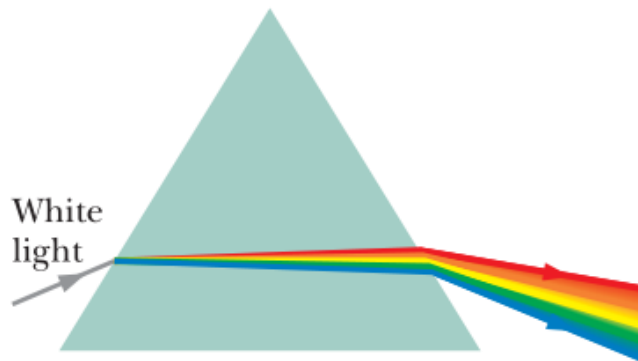


If the incident beam is oriented as shown, total internal reflection occurs on the  $45^\circ$  faces (because, for a glass–air interface,  $\theta_{\text{crit}} = 41.1^\circ$ ).



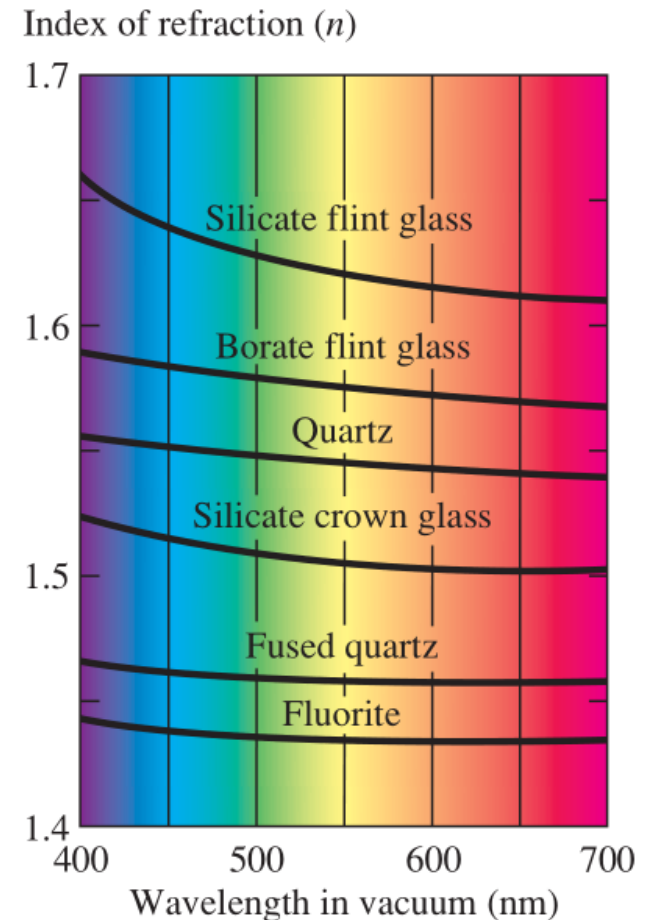
# 33-4 Dispersion

The refractive index of most media increases with the frequency of light waves. Therefore, when a beam of white light enters a prism, violet light is deflected the most, followed by red light, which is deflected the least, forming a color dispersion spectrum of various colors.



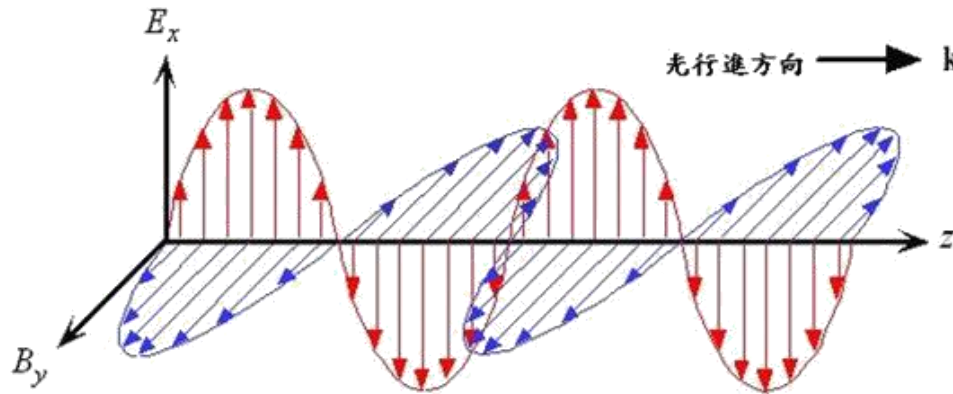
(b)

**Figure 33-20** (a) A triangular prism separating white light into its component colors. (b) Chromatic dispersion occurs at the first surface and is increased at the second surface.

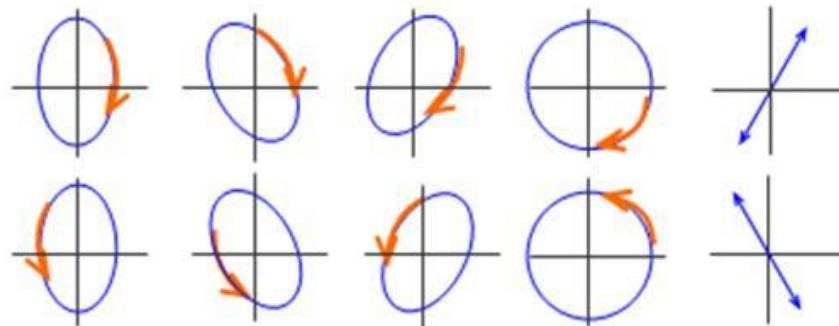


# 33-5 Polarization

**Polarization** is a property of **transverse waves**, and electromagnetic waves are transverse waves.



The direction of polarization is determined by the direction of the electric field intensity. When the electric field vector vibrates along a straight line, it is called linearly polarized; when the trajectory of the end of the electric field vector is a circle or an ellipse, it is called circular polarization or elliptical polarization.



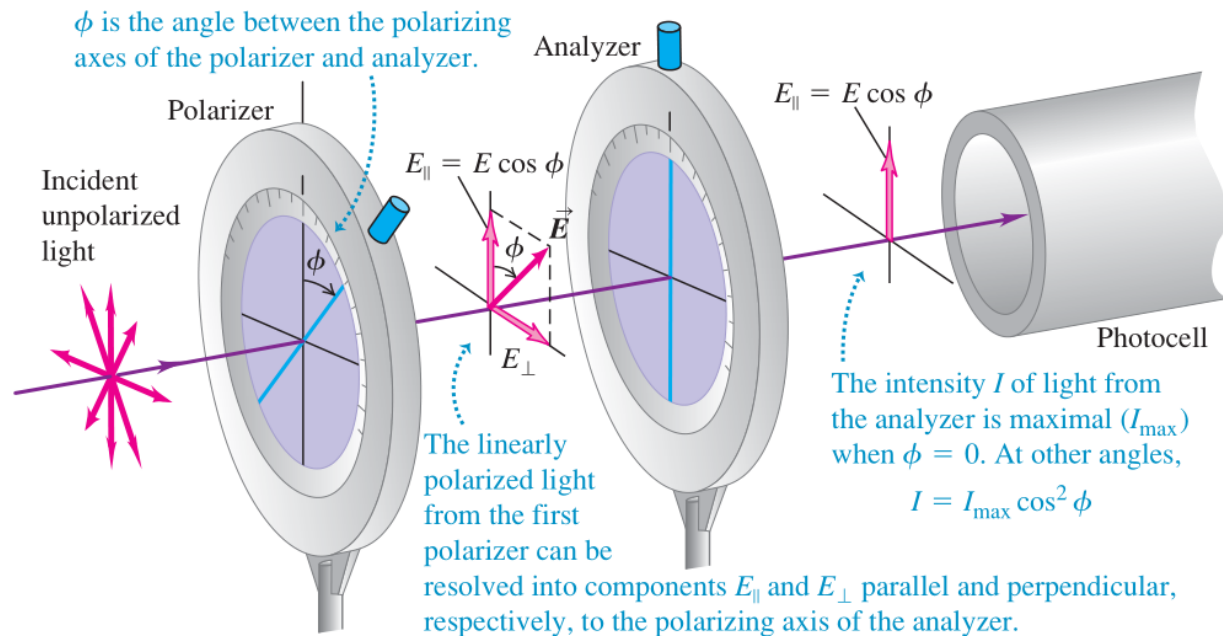
# 33-5 Polarization

**Natural light** is the unpolarized light, which is composed of linearly polarized light in various directions.

Natural light becomes linearly polarized light after passing through a **polarizing filter**.

- The intensity of linearly polarized light after passing through a polarizer is:

$$I = I_{\max} \cos^2 \phi \quad \text{——} \textit{Malus's Law}$$

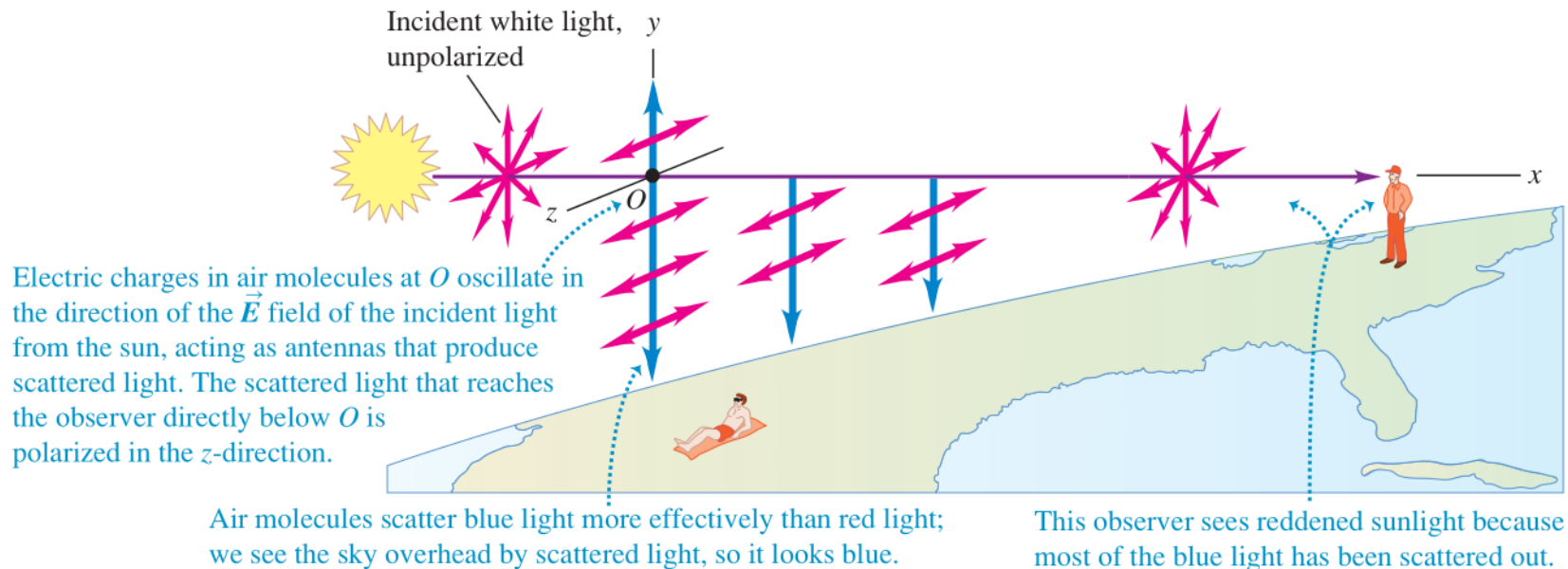


## 33-6 Scattering of Light

When air molecules absorb sunlight, they radiate in all directions; this process is called **scattering**.

Scattered light is **linearly polarized**, and it can be proven that the intensity of scattered light is proportional to the fourth power of the frequency. Therefore, the intensity of blue light in scattered light is approximately nine times that of red light, which is why the sky is blue.

**33.32** When the sunbathing observer on the left looks up, he sees blue, polarized sunlight that has been scattered by air molecules. The observer on the right sees reddened, unpolarized light when he looks at the sun.



## 33-7 Huygens' Principle

---

The laws of reflection and refraction of light can be derived from **Huygens' principle using the wave theory of light**. This principle was proposed in 1678 by the Dutch scientist Christiaan.

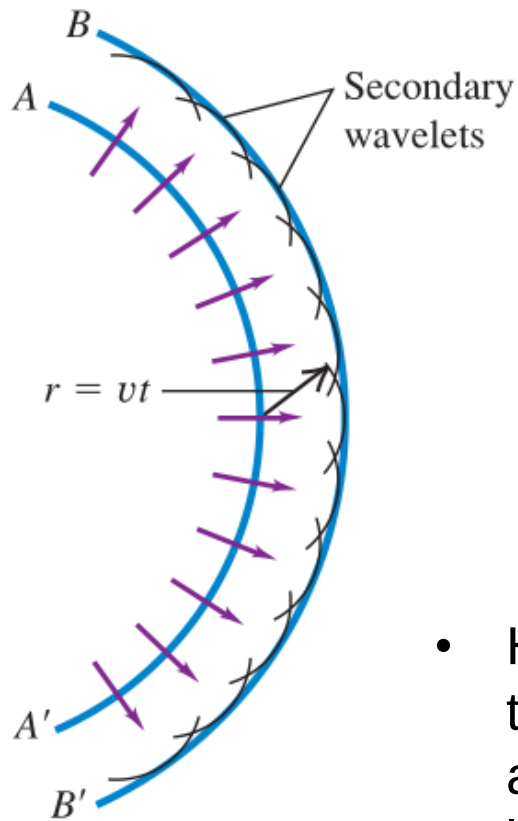
Huygens assumed **that every point of a wave front may be considered the source of secondary wavelets that spread out in all directions with a speed equal to the speed of propagation of the wave**. The new wave front at a later time is then found by constructing a *surface tangent* to the secondary wavelets or, as it is called, the *envelope of the wavelets*.

All the results that we obtain from Huygens's principle can also be obtained from Maxwell's equations. Thus it is not an independent principle, but it is often very convenient for calculations with wave phenomena.



# 33-7 Huygens' Principle

**33.34** Applying Huygens's principle to wave front  $AA'$  to construct a new wave front  $BB'$ .



- Each point on the wavefront can be considered as a source of secondary waves, emitting secondary waves in all directions at a certain wave speed; The new wavefront is the envelope of the wavelets.
- Huygens's principle can be used to derive the laws of reflection and refraction. Also to analyze the diffraction and interference of light.

